Chapter 2. Permutations, Cosets, and Direct Products

Permutation groups, Cayley, cycles and the alternating group, Lagrange, direct products, finitely generated abelian groups, plane isometries

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2.1 Groups of permutations (§8)

Motivation. Groups first arose as permutations of the roots of a polynomial, and Cayley's theorem says nothing is lost by thinking of groups that way: every group is a group of permutations. Permutation groups therefore give concrete models for all groups.

A **permutation** of a set A is a bijection $A \to A$; their product $\sigma\tau$ is the composition $\sigma \circ \tau$ (apply τ first).

Theorem 2.1 (8.5)

The permutations of a nonempty set A form a group S_A under composition.

Proof. A composite of bijections is a bijection, composition is associative (Theorem 1.2), the identity map is an identity, and the inverse function of a bijection is an inverse. ■

$S_n = S_{\{1,\dots,n\}}$ is the **symmetric group**, of order $n!$; it is nonabelian for $n \geq 3$. The **dihedral group** D_n ($n \geq 3$) is the group of symmetries of a regular n-gon, a subgroup of S_n of order $2n$; $D_3 = S_3$.

Lemma 2.2 (8.15)

If $\phi : G \to G'$ is one-to-one and $\phi(xy) = \phi(x)\phi(y)$ for all x, y, then $\phi[G]$ is a subgroup of G' and ϕ is an isomorphism of G with $\phi[G]$.

Proof. $\phi(x)\phi(y) = \phi(xy) \in \phi[G]$. From $\phi(e)\phi(e) = \phi(e) = e'\phi(e)$, cancellation gives $\phi(e) = e'$; then $\phi(a)\phi(a^{-1}) = \phi(e) = e'$ gives $\phi(a)^{-1} = \phi(a^{-1}) \in \phi[G]$. So $\phi[G] \leq G'$ (Theorem 1.8), and $\phi : G \to \phi[G]$ is a bijection preserving the operation. ■

Theorem 2.3 (8.16, Cayley's theorem)

Every group is isomorphic to a group of permutations.

Proof. For $x \in G$ let $\lambda_x : G \to G$, $\lambda_x(g) = xg$. It is one-to-one ($xg_1 = xg_2 \Rightarrow g_1 = g_2$) and onto ($\lambda_x(x^{-1}c) = c$), so $\lambda_x \in S_G$. The map $\phi(x) = \lambda_x$ is one-to-one ($\lambda_x(e) = x$) and $\lambda_{xy}(g) = xyg = \lambda_x(\lambda_y(g))$, so $\phi(xy) = \phi(x)\phi(y)$. Apply Lemma 2.2. ■

ϕ is the **left regular representation**; with $\rho_x(g) = gx^{-1}$ one gets the right regular representation.

2.2 Orbits, cycles, and the alternating groups (§9)

Motivation. To compute with permutations we need a normal form. Every permutation splits into disjoint cycles (its orbits), and every cycle into transpositions; the surprising fact that the parity of the number of transpositions is well defined produces the alternating group A_n, which will turn out to be simple for $n \geq 5$, the group-theoretic reason the quintic is not solvable by radicals.

For $\sigma \in S_A$, $a \sim b$ iff $b = \sigma^n(a)$ for some $n \in \mathbb{Z}$ is an equivalence relation; its cells are the **orbits** of σ. A **cycle** is a permutation with at most one orbit of more than one element; $(a_1, a_2, \dots, a_m)$ sends $a_i \mapsto a_{i+1}$, $a_m \mapsto a_1$, its **length** is m; a cycle of length 2 is a **transposition**.

Theorem 2.4 (9.8)

Every permutation of a finite set is a product of disjoint cycles.

Proof. Let $B_1, \dots, B_r$ be the orbits of σ and μ_i the cycle that agrees with σ on B_i and is the identity elsewhere (on an orbit $B = \{b, \sigma(b), \dots, \sigma^{m-1}(b)\}$, σ acts as the cycle $(b, \sigma(b), \dots, \sigma^{m-1}(b))$). The orbits are disjoint, so $\sigma = \mu_1\mu_2\cdots\mu_r$: each x lies in exactly one B_i, and only μ_i moves it. ■

Disjoint cycles commute, and the decomposition is unique up to order (the cycles are determined by the orbits).

Corollary 2.5 (9.12)

Any permutation of a finite set with at least two elements is a product of transpositions.

Proof. $(a_1, a_2, \dots, a_m) = (a_1, a_m)(a_1, a_{m-1})\cdots(a_1, a_2)$, and the identity is $(1,2)(1,2)$. ■

Theorem 2.6 (9.15)

No permutation in S_n is both a product of an even number and of an odd number of transpositions.

Proof. (Orbit counting.) Let $\sigma \in S_n$ have r orbits (counting fixed points) and set $N(\sigma) = n - r$. For a transposition $\tau = (i, j)$, $N(\tau\sigma) = N(\sigma) \pm 1$: if i, j lie in the same orbit of σ, write the cycle as $(i, b_1, \dots, b_h, j, c_1, \dots, c_k)$; then (i, j) times it is $(i, c_1, \dots, c_k)(j, b_1, \dots, b_h)$, so the number of orbits increases by 1. If i, j lie in different orbits, the same computation read backwards shows the two orbits merge, decreasing the count by 1. Since $N(\mathrm{id}) = 0$, a product of k transpositions has $N \equiv k \pmod 2$. So the parity of k is determined by σ. ■

A permutation is **even** or **odd** according to this parity; a cycle of length m has parity of $m - 1$.

Theorem 2.7 (9.20)

For $n \geq 2$, the even permutations of $\{1, \dots, n\}$ form a subgroup A_n of S_n of order $n!/2$.

Proof. Products of even permutations are even, the identity is even, and the inverse of $\tau_1 \cdots \tau_k$ is $\tau_k \cdots \tau_1$; so $A_n \leq S_n$. Fix $\tau = (1,2)$. $\lambda_\tau(\sigma) = \tau\sigma$ maps A_n into the odd permutations B_n and B_n into A_n, and $\lambda_\tau^2 = \mathrm{id}$; so it is a bijection $A_n \to B_n$ and $|A_n| = |B_n| = n!/2$. ■

A_n is the **alternating group**.

2.3 Cosets and the theorem of Lagrange (§10)

Motivation. How large can a subgroup be? Cosets of H chop G into equal-sized pieces, so $|H|$ must divide $|G|$. This single counting argument already forces strong conclusions: groups of prime order are cyclic, and $a^{|G|} = e$, which contains Fermat's little theorem.

Theorem 2.8 (10.1)

Let $H \leq G$. The relations $a \sim_L b \iff a^{-1}b \in H$ and $a \sim_R b \iff ab^{-1} \in H$ are equivalence relations on G.

Proof. For $\sim_L$: $a^{-1}a = e \in H$; if $a^{-1}b \in H$ then $(a^{-1}b)^{-1} = b^{-1}a \in H$; if $a^{-1}b, b^{-1}c \in H$ then their product $a^{-1}c \in H$. $\sim_R$ is analogous. ■

The cell of a under $\sim_L$ is the **left coset** $aH = \{ah : h \in H\}$ (as $a^{-1}b = h \iff b = ah$); under $\sim_R$ it is the **right coset** Ha. For abelian G they coincide.

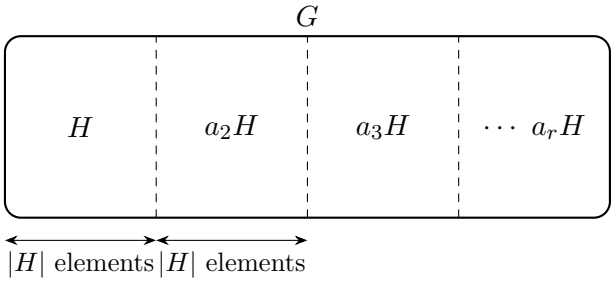


Figure. *The left cosets of H partition G into $r = (G : H)$ blocks of equal size $|H|$, so $|G| = r\,|H|$.*

Theorem 2.9 (10.10, Lagrange)

If H is a subgroup of a finite group G, then $|H|$ divides $|G|$.

Proof. $h \mapsto ah$ is a bijection $H \to aH$ (one-to-one by cancellation), so every left coset has $|H|$ elements. The left cosets partition G (Theorems 2.8, 1.1); if there are r of them, $|G| = r|H|$. ■

Corollary 2.10 (10.11)

Every group of prime order is cyclic.

Proof. For $a \neq e$, $|\langle a\rangle| > 1$ divides p, so equals p and $\langle a\rangle = G$. ■

Theorem 2.11 (10.12)

The order of an element of a finite group divides the order of the group.

Proof. It is the order of the cyclic subgroup it generates. ■

The number of left cosets of H in G is the **index** $(G : H)$; the map $aH \mapsto Ha^{-1}$ is a bijection between left and right cosets, so the index is the same for both.

Theorem 2.12 (10.14)

If $K \leq H \leq G$ with $(G : H)$ and $(H : K)$ finite, then $(G : K)$ is finite and $(G : K) = (G : H)(H : K)$.

Proof. Let $G = \bigsqcup_{i\leq r} a_iH$ and $H = \bigsqcup_{j\leq s} b_jK$. Then $a_iH = \bigsqcup_j a_ib_jK$, so $G = \bigcup_{i,j} a_ib_jK$. These rs cosets are distinct: if $a_ib_jK = a_pb_qK$, then since $b_jK, b_qK \subset H$, the cosets a_iH and a_pH meet, so $i = p$; then $b_jK = b_qK$ and $j = q$. ■

The converse of Lagrange fails: A_4 (order 12) has no subgroup of order 6.

2.4 Direct products and finitely generated abelian groups (§11)

Motivation. Direct products build new groups from old; the main theorem of the section goes backwards and says that every finitely generated abelian group is assembled in this way from cyclic groups. Abelian groups are therefore completely classified by a list of integers, which lets us enumerate, for instance, all abelian groups of order 360.

Theorem 2.13 (11.2)

For groups $G_1, \ldots, G_n$, $\prod G_i$ is a group (the **direct product**) under $(a_1, \ldots, a_n)(b_1, \ldots, b_n) = (a_1 b_1, \ldots, a_n b_n)$.

Proof. The operation is componentwise, so associativity holds componentwise; $(e_1, \ldots, e_n)$ is the identity and $(a_1^{-1}, \ldots, a_n^{-1})$ the inverse. ■

Theorem 2.14 (11.5)

$\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic, and isomorphic to $\mathbb{Z}_{mn}$, iff $\gcd(m, n) = 1$.

Proof. If $\gcd(m, n) = 1$, the order of $(1, 1)$ is the least k divisible by both m and n, namely mn, so $(1, 1)$ generates. If $d = \gcd(m, n) > 1$, then mn/d is divisible by m and n, so every element (r, s) satisfies $(mn/d)(r, s) = (0, 0)$; no element has order mn. ■

Corollary 2.15 (11.6)

$\prod_{i=1}^{n} \mathbb{Z}_{m_i}$ is cyclic and isomorphic to $\mathbb{Z}_{m_1 \cdots m_n}$ iff the m_i are pairwise relatively prime.

Proof. Induction using Theorem 2.14 and $\prod_{i \le n} \mathbb{Z}_{m_i} \cong (\prod_{i<n} \mathbb{Z}_{m_i}) \times \mathbb{Z}_{m_n}$: if pairwise relatively prime, $m_1 \cdots m_{n-1}$ and m_n are relatively prime. Conversely if $\gcd(m_i, m_j) > 1$, the subgroup $\mathbb{Z}_{m_i} \times \mathbb{Z}_{m_j}$ is not cyclic, but subgroups of cyclic groups are cyclic. ■

Theorem 2.16 (11.9)

If $a_i \in G_i$ has finite order r_i, then $(a_1, \ldots, a_n)$ has order $\operatorname{lcm}(r_1, \ldots, r_n)$.

Proof. $(a_1, \ldots, a_n)^k = (e_1, \ldots, e_n)$ iff $r_i \mid k$ for all i; the least such $k > 0$ is the lcm. ■

Theorem 2.17 (11.12, fundamental theorem of finitely generated abelian groups)

Every finitely generated abelian group is isomorphic to

$$\mathbb{Z}_{(p_1)^{r_1}} \times \mathbb{Z}_{(p_2)^{r_2}} \times \cdots \times \mathbb{Z}_{(p_n)^{r_n}} \times \mathbb{Z} \times \cdots \times \mathbb{Z},$$

with primes p_i (not necessarily distinct) and $r_i \in \mathbb{Z}^+$. The direct product is unique except for the order of the factors; the number of factors $\mathbb{Z}$ is the **Betti number**.

Proof [Supplement].

Step 1. (Stacked bases.) Write G additively, generated by $x_1, \ldots, x_n$. The map $\mathbb{Z}^n \to G$, $(m_i) \mapsto \sum m_i x_i$, is an onto homomorphism with kernel K, and $G \cong \mathbb{Z}^n / K$ (Theorem 3.9). We show: there are a basis $f_1, \ldots, f_n$ of $\mathbb{Z}^n$ and integers $d_1 \mid d_2 \mid \cdots \mid d_k$ $(k \le n,\ d_i > 0)$ such that $d_1 f_1, \ldots, d_k f_k$ is a basis of K. Induction on n; if $K = 0$ take $k = 0$. Over all bases $f_1, \ldots, f_n$ of $\mathbb{Z}^n$ and all nonzero $v = \sum c_i f_i \in K$, choose one in which some coefficient has the least possible positive absolute value d; renumbering and changing signs, $c_1 = d$.

(i) d divides every c_j: if $c_j = qd + r$ with $0 < r < d$, the basis $f_1 + q f_j, f_2, \ldots, f_n$ (still a basis, as $f_1 = (f_1 + q f_j) - q f_j$) gives v the j-th coefficient $c_j - qd = r$, contradicting minimality. So $v = d g_1$ with $g_1 = f_1 + \sum_{i>1} (c_i/d) f_i$, and $g_1, f_2, \ldots, f_n$ is a basis.

(ii) For $w = a g_1 + \sum_{i>1} b_i f_i \in K$, d divides a (else $w - qv$ has coefficient $a - qd \in (0, d)$), and d divides every b_i (apply (i) to the element $v + (w - \frac{a}{d} v) = d g_1 + \sum b_i f_i$ of K). Hence $K = \mathbb{Z}\, d g_1 \oplus K'$ with $K' = K \cap \langle f_2, \ldots, f_n \rangle$, and d divides all coefficients of elements of K'.

(iii) By induction, $\langle f_2, \dots, f_n \rangle \cong \mathbb{Z}^{n-1}$ has a basis $g_2, \dots, g_n$ with $d_2 g_2, \dots, d_k g_k$ a basis of K' and $d_2 \mid \cdots \mid d_k$. The coordinates of g_2 with respect to $f_2, \dots, f_n$ have gcd 1 (it is part of a basis), and d divides all coordinates of $d_2 g_2$, so $d \mid d_2$. Put $d_1 = d$.

Step 2. (Existence.) Then $G \cong \mathbb{Z}^n / K \cong \mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_k} \times \mathbb{Z}^{n-k}$, via $\sum m_i f_i \mapsto (m_1 \bmod d_1, \dots, m_k \bmod d_k, m_{k+1}, \dots, m_n)$, an onto homomorphism with kernel K. Drop factors with $d_i = 1$, and split each $\mathbb{Z}_d$ with $d = \prod p_j^{s_j}$ as $\prod \mathbb{Z}_{p_j^{s_j}}$ (Corollary 2.15).

Step 3. (Uniqueness.) The torsion subgroup T (elements of finite order) corresponds to the finite factors, and $G/T \cong \mathbb{Z}^b$, where b is the number of $\mathbb{Z}$ factors; b is determined since $\mathbb{Z}^b / 2\mathbb{Z}^b$ has 2^b elements. For the finite part and a prime p, let T_p be the elements of p-power order, the product of the factors $\mathbb{Z}_{p^r}$ with this p. If $T_p \cong \prod_j \mathbb{Z}_{p^{r_j}}$, then $p^{k-1}T_p / p^k T_p$ has order p^{N_k}, where N_k is the number of j with $r_j \geq k$. Since these groups depend only on G, the multiset $\{r_j\}$ is determined. ■

Example 2.1

Abelian groups of order $360 = 2^3 \cdot 3^2 \cdot 5$, up to isomorphism: $3 \cdot 2 \cdot 1 = 6$ of them (partitions of 3 for the 2-part, of 2 for the 3-part), e.g. $\mathbb{Z}_8 \times \mathbb{Z}_9 \times \mathbb{Z}_5$ and $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$.

A group is **decomposable** if it is isomorphic to a direct product of two proper nontrivial subgroups.

Theorem 2.18 (11.15)

The finite indecomposable abelian groups are exactly the cyclic groups of prime power order.

Proof. By Theorem 2.17 a finite indecomposable abelian group has one factor $\mathbb{Z}_{p^r}$. Conversely, if $\mathbb{Z}_{p^r} \cong H \times K$ with nontrivial H, K of orders p^s, p^t $(s, t < r)$, then every element has order at most $p^{\max(s,t)} < p^r$, contradicting cyclicity. ■

Theorem 2.19 (11.16)

If m divides the order of a finite abelian group G, then G has a subgroup of order m.

Proof. Write $G \cong \prod \mathbb{Z}_{(p_i)^{r_i}}$ and $m = \prod_p p^{t_p}$ with $t_p \leq \sum_{p_i = p} r_i$. For each prime p choose $s_i \leq r_i$ (for $p_i = p$) with $\sum s_i = t_p$. The subgroup $\langle (p_i)^{r_i - s_i} \rangle \leq \mathbb{Z}_{(p_i)^{r_i}}$ has order $(p_i)^{s_i}$, and the product of these subgroups has order m. ■

Theorem 2.20 (11.17)

If m is square free, every abelian group of order m is cyclic.

Proof. In Theorem 2.17 the primes p_i must be distinct with $r_i = 1$; then $\prod \mathbb{Z}_{p_i} \cong \mathbb{Z}_m$ by Corollary 2.15. ■

2.5 Plane isometries (§12)

Motivation. The symmetries of the plane form a group, and classifying them gives a geometric illustration of the abstract theory: each isometry is a translation, rotation, reflection or glide reflection, and the possible symmetry groups of plane patterns are severely restricted.

An **isometry** of $\mathbb{R}^2$ is a distance-preserving map. Every isometry is a translation, a rotation, a reflection, or a glide reflection; it is affine, $\phi(x) = Ax + c$ with A orthogonal (an isometry fixing

0 preserves norms, hence inner products by polarization, hence is linear). Orientation-preserving isometries ($\det A = 1$) are translations and rotations.

Theorem 2.21 (12.5)

Every finite group G of isometries of the plane is isomorphic to $\mathbb{Z}_n$ or to a dihedral group D_n.

Proof.

Step 1. Let $G = \{\phi_1, \dots, \phi_m\}$ and $P = \frac{1}{m}\sum_i \phi_i(0)$. Since each ϕ is affine, $\phi(P) = \frac{1}{m}\sum_i \phi(\phi_i(0)) = \frac{1}{m}\sum_i \phi_i(0) = P$ ($\phi\phi_i$ runs over G). Translating, assume $P = 0$; then G consists of rotations about 0 and reflections in lines through 0.

Step 2. The rotations form a subgroup H (a product of two orientation-preserving isometries is one). If $H \neq \{\mathrm{id}\}$, let ρ be the rotation in H by the least positive angle θ. Every rotation in H, by angle $\alpha \in [0, 2\pi)$, satisfies $\alpha = k\theta + \beta$ with $0 \leq \beta < \theta$, and ρ^{-k} composed with it rotates by β, so $\beta = 0$; hence $H = \langle\rho\rangle \cong \mathbb{Z}_n$.

Step 3. If $G = H$ we are done. Otherwise let $\mu \in G$ be a reflection. For any reflection $\mu' \in G$, $\mu^{-1}\mu'$ preserves orientation, so lies in H; hence $G = H \cup \mu H$ has order $2n$. Since $\mu\rho\mu^{-1} = \rho^{-1}$, G is generated by ρ (order n) and μ (order 2) with this relation, which is the dihedral group D_n (for $n = 1$, $\mathbb{Z}_2$; for $n = 2$, the Klein group D_2). ■

2.6 Selected exercises

Motivation. These exercises from the book use the tools of this chapter: conjugating a transposition, parity of permutations, and Lagrange's theorem. Together they show that S_n has trivial center and two generators, and that groups without proper nontrivial subgroups are exactly the groups of prime order.

Exercise 2.1 (§8, Ex. 47; the center of S_n)

Let $n \geq 3$. Show that the only $\sigma \in S_n$ with $\sigma\gamma = \gamma\sigma$ for all $\gamma \in S_n$ is the identity ι.

Solution. Suppose $\sigma \neq \iota$, say $\sigma(i) = j$ with $j \neq i$. Since $n \geq 3$ there is $k \notin \{i, j\}$; let $\gamma = (j, k)$. As γ fixes i,

$$(\sigma\gamma)(i) = \sigma(i) = j, \qquad (\gamma\sigma)(i) = \gamma(j) = k \neq j.$$

So $\sigma\gamma \neq \gamma\sigma$. In particular S_n is nonabelian for $n \geq 3$. ■

Exercise 2.2 (§9, Ex. 29; even permutations in a subgroup)

Let $n \geq 2$ and $H \leq S_n$. Show that either all permutations in H are even, or exactly half of them are even.

Solution. Let E and O be the sets of even and odd permutations in H, and suppose $O \neq \emptyset$; fix $\tau \in O$.

Step 1. Writing permutations as products of transpositions and concatenating, the product of an odd and an even permutation is odd and the product of two odd ones is even (parity is well defined by Theorem 2.6). The inverse of $\tau_1 \cdots \tau_k$ is $\tau_k \cdots \tau_1$, so τ^{-1} is odd.

Step 2. Define $\lambda : E \to O$, $\lambda(\sigma) = \tau\sigma$. It maps into O by Step 1 and since H is closed. It is one-to-one by cancellation. It is onto: for $\rho \in O$, $\tau^{-1}\rho \in H$ is even by Step 1 and $\lambda(\tau^{-1}\rho) = \rho$.

Step 3. Hence $|E| = |O|$ and $|E| = |H|/2$. ■

Exercise 2.3 (§9, Ex. 39; two generators for S_n)

Show that S_n $(n \geq 2)$ is generated by $\tau = (1, 2)$ and $\rho = (1, 2, 3, \ldots, n)$.

Solution. Let K be the subgroup generated by τ and ρ.

Step 1. (Conjugating a transposition.) For $\sigma \in S_n$ and $a \neq b$, $\sigma(a, b)\sigma^{-1} = (\sigma(a), \sigma(b))$. Indeed, $\sigma(a, b)\sigma^{-1}$ sends $\sigma(a) \mapsto \sigma(b)$ and $\sigma(b) \mapsto \sigma(a)$, and if $x \notin \{\sigma(a), \sigma(b)\}$, then $\sigma^{-1}(x) \notin \{a, b\}$ is fixed by (a, b), so $x \mapsto x$.

Step 2. (Adjacent transpositions.) For $0 \leq r \leq n - 2$, $\rho^r(1) = r + 1$ and $\rho^r(2) = r + 2$, so Step 1 gives $\rho^r \tau \rho^{-r} = (r + 1, r + 2) \in K$. Thus $(i, i + 1) \in K$ for $1 \leq i \leq n - 1$.

Step 3. (All transpositions.) We show $(i, j) \in K$ for $i < j$ by induction on $j - i$; $j - i = 1$ is Step 2. If $j - i \geq 2$, then $(i, j - 1) \in K$ by induction, and Step 1 with $\sigma = (j - 1, j)$, which fixes i and sends $j - 1$ to j, gives

$$(j - 1, j)(i, j - 1)(j - 1, j)^{-1} = (i, j) \in K.$$

Step 4. Every permutation is a product of transpositions (Corollary 2.5), so $K = S_n$. ■

Exercise 2.4 (§10, Ex. 37; groups without proper subgroups)

Show that a group G with at least two elements but no proper nontrivial subgroups is finite of prime order.

Solution.

Step 1. Choose $a \neq e$. Then $\langle a \rangle \neq \{e\}$, so $\langle a \rangle = G$ and G is cyclic.

Step 2. G is finite. Otherwise a has infinite order, so $a^2 \neq e$ and $\langle a^2 \rangle \neq \{e\}$. Also $a \notin \langle a^2 \rangle$, since $a = a^{2k}$ would give $a^{2k-1} = e$ with $2k - 1 \neq 0$. So $\langle a^2 \rangle$ is a proper nontrivial subgroup, a contradiction.

Step 3. Let $|G| = n \geq 2$. If n were not prime, write $n = dm$ with $1 < d < n$, so $1 < m < n$. By Theorem 1.15, a^d has order $n/\gcd(n, d) = n/d = m$, so $\langle a^d \rangle$ is a proper nontrivial subgroup, a contradiction. Hence n is prime (and $G \cong \mathbb{Z}_n$ by Corollary 2.10). ■